Mechanical Engineering Experiments (II) — Report 2

機械工程實驗(二) 第二次報告

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Q1:

$FPS = 50.003\ frames/s$

Time conversion: $t_i = \frac{i}{FPS},\ \ i = 0, 1, 2, 3 \ldots, N-1$

$Pixels = 948.76,\ True\ Length = 0.14m$

Position conversion: $\begin{cases} x_m = x_{pixel} \times \frac{True\ Length}{Pixels} = 0.0001476x_{pixel} \\ y_m = y_{pixel} \times \frac{True\ Length}{Pixels} = 0.0001476y_{pixel} \end{cases}$

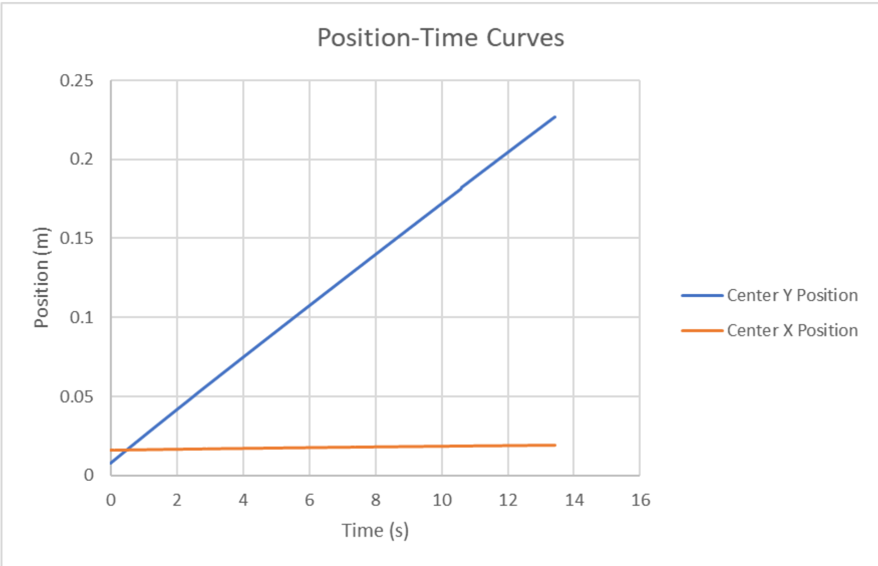

Figure1. Position-Time Curves for center of X and Y

Q2:

(a) .

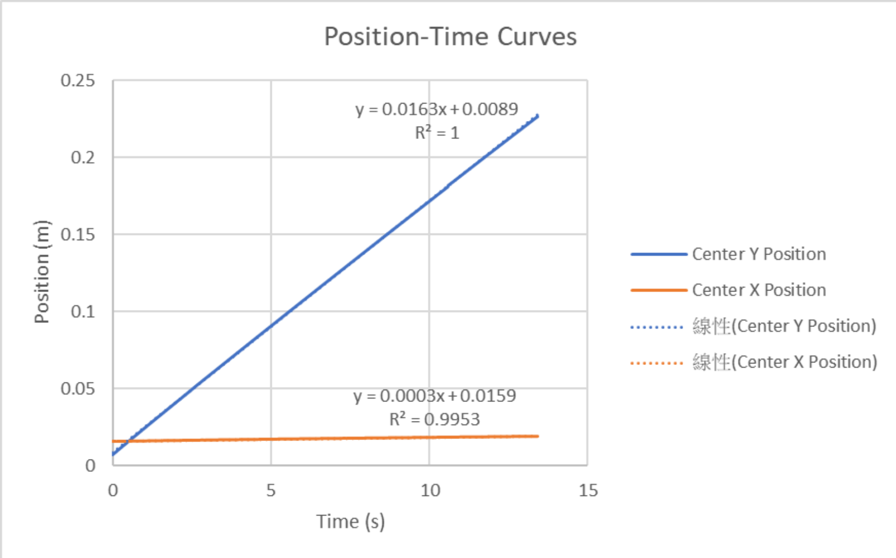

Figure2. Position-Time Curves for center of X and Y with linear regression

Falling speed along X dir.:

$X(t) = 0.0003t + 0.0159,\ U_X = \frac{dX}{dt} = 0.0003\ m/s$

Falling speed along Y dir.:

$X(t) = 0.0163t + 0.0089,\ U_Y = \frac{dY}{dt} = 0.0163\ m/s$

For the pick-out criterion, by observing the plots above, there are no significant fluctuation. And also due to the high linearity of two sets of data,

$\begin{cases} R_X^2 = 0.9953, R_X \approx 0.998 > 0.99 \\ R_Y^2 = 1.000, R_Y = 1.000 > 0.99 \end{cases}$, we don't need to pick out any unreasonable data.

(b) .

Judging Criterion: If the velocity ratio along difference axis is smaller than 3%, i.e. $\frac{U_X}{U_Y} < 3\%$, we can ignore the X axis velocity during the latter discussion.

For our experiment data, $Velocity\ ratio = \frac{U_X}{U_Y} = \frac{0.0003}{0.0163} \approx 1.840\% < 3\%$

→ velocity along X axis can be ignored

If the falling speed reaches its terminal magnitude, acceleration along the axis $a = \frac{dv}{dt} \approx 0$. By the plot we obtain in Q1(b), the velocity of Y direction shows perfect linearity (R ≈ 1) without any obvious curve changes, which means in the Y direction, $a_Y = \frac{dv_Y}{dt} \approx 0$.

→ ball's velocity along Y axis had reached its terminal magnitude

Q3:

Force balance equation: $\frac{4}{3}\pi r^3 \rho_{ball} g = 6\pi\mu r u + \frac{4}{3}\pi r^3 \rho_{liquid} g$

Known data (in SI):

$r = \frac{d}{2} = \frac{4}{2} = 2\ mm = 0.002\ m$

$V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi \times 0.002^3 \approx 3.351 \times 10^{-8}\ m^3$

$m = 0.08g$

$\rho_{ball} = \frac{m}{V} = \frac{0.08}{3.351\times10^{-8}} \approx 2.387 \times 10^6 g/m^3 = 2387\ kg/m^3$

$g = 9.81\ m/s^2$

$u = 0.0163\ m/s$ (u_X can be ignored)

$\rho_{liquid} = 1258\ kg/m^3$

→ $\mu_{exp} = \frac{\frac{4}{3}\pi r^3(\rho_{ball}-\rho_{liquid})g}{6\pi r u} = \frac{\frac{4}{3}\pi\times0.002^3(2387-1258)\times9.81}{6\pi\times0.002\times0.0163} \approx 0.604\ Pa\cdot s$

$\mu_{viscometer} = 0.440 Pa\cdot s$

$Er = \frac{\mu_{exp}-\mu_{viscometer}}{\mu_{viscometer}} = \frac{0.604-0.440}{0.44} = 37.273\%$

Q4:

Experiment estimation:

Known data (in SI):

$chute\ area\ A = 0.1 \times 0.1 = 0.01\ m^2$

$$\begin{cases} flow\ rate_{observing} = 34.3\ L/min = 5.72 \times 10^{-4}\ m^3/s \\ flow\ rate_{PIV} = 60\ L/min = 1 \times 10^{-3}\ m^3/s \end{cases}$$

$$\begin{cases} U_{observing} = \frac{flow\ rate_{observing}}{A} = 0.0572\ m/s \\ U_{PIV} = \frac{flow\ rate_{PIV}}{A} = 0.1\ m/s \end{cases}$$

$D_h = 0.1m$

$$\nu_{water@25℃} = \frac{\mu}{\rho} = \frac{0.00089}{997} \approx 8.93 \times 10^{-7}\ m^2/s$$

$$\rightarrow \begin{cases} Re_{observing} = \frac{U_{observing} D_h}{\nu} = \frac{0.0572 \times 0.1}{8.93 \times 10^{-7}} \approx 6405.375 \\ Re_{PIV} = \frac{U_{PIV} D_h}{\nu} = \frac{0.1 \times 0.1}{8.93 \times 10^{-7}} \approx 11198.208 \end{cases}$$

$$\begin{cases} f_{observing} \approx 0.629\ Hz\ \ (6.5\ vortex\ pairs\ for\ 10.33\ sec.) \\ f_{PIV} \approx 1.380\ Hz\ \ (2\ vortex\ pairs\ for\ 145\ frame\ (1.45\ sec.)) \end{cases}$$

$L = 0.02m$

$$\rightarrow \begin{cases} St_{observing,exp} = \frac{f_{observing} L}{U_{observing}} = \frac{0.629 \times 0.02}{0.0572} \approx 0.2199 \\ St_{PIV,exp} = \frac{f_{PIV} L}{U_{PIV}} = \frac{1.380 \times 0.02}{0.1} \approx 0.2760 \end{cases}$$

Theoretical estimation:

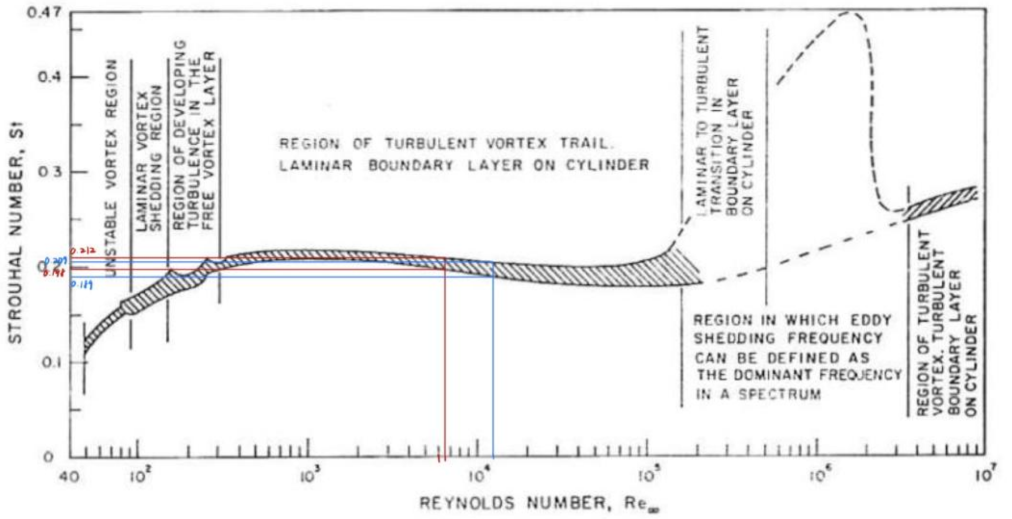

Figure3. Relationship between the Strouhal number and the Reynolds number of a cylinder

For observing ($Re_{observing} \approx 6405.375$): $St_{observing,theoretical} = 0.198\ (rough - surfaced) \sim 0.212\ (smooth - surfaced)$

For PIV ($Re_{PIV} \approx 6405.375$): $St_{PIV,theoretical} = 0.189\ (rough - surfaced) \sim 0.207\ (smooth - surfaced)$

Error Analysis:

$$Error_{observing} = \frac{St_{observing,exp} - St_{observing,theoretical}}{St_{observing,theoretical}} = \frac{0.2199-0.212}{0.212} \approx 3.726\%$$

$$Error_{PIV} = \frac{St_{PIV,exp} - St_{PIV,theoretical}}{St_{PIV,theoretical}} = \frac{0.2760-0.207}{0.207} \approx 33.3\%$$

For the error analysis, we can see that observing shedding frequency provides a better approach to the theoretical estimation with only 3.726% error over the upper limit. While PIV has a large error of 33.3% over the upper limit.

Q5:

(a) .

$Cylinder\ diameter\ D = 4\ mm$

$D_{error} = 0.05 \times 4 = 0.2\ mm$

However, the distance we focus on is the absolute length between real center and estimated center, which is half of diameter error.

$L_{er} = \frac{1}{2} D_{error} = 0.1\ mm$

(b) .

From Q2(a). → $U_Y = 0.0163\ m/s$

FPS = 50.003fps

$Falling\ distance\ d = \frac{U_Y}{FPS} = \frac{0.0163}{50.003} \approx 0.000326\ m = 0.326\ mm$

(c) .

Based on the previous results: $d = 0.326\ mm > L_{er} = 0.1\ mm$, which means the ball's falling distance between two consecutive frames is larger than the location error. We can trust the estimated falling velocity.

If the falling distance is smaller than the location error, the estimated falling velocity calculation would be dominated by the location error. For example, if the true moving length between two consecutive frames is only 0.05mm, but the error (e.g. 0.1mm) is larger than this value, we can't distinguish whether the calculation result is just an error fluctuation or the true moving distance we really focus on. In conclusion, if $L_{er} > d$, the data will be loss of credibility.

(d) .

If the FPS applied to calculate the falling velocity is too high, the time interval between each consecutive frame Δt would be too small. While

$Falling\ distance\ d = \frac{U_Y}{FPS}$, if $\uparrow FPS\ \rightarrow \downarrow d$. However, the error between the estimated center and real center is all exist and consistent due to the instrument problem. According to the discussion in Q5(d), if FPS is large enough or reaches almost same magnitude as d to make $L_{er} > d$, the data will be loss of credibility.